

GOOGLE CYBERSECURITY CERTIFICATE

Incident Response

Google Cybersecurity Professional Certificate · Study Guide

Three incident-response writeups spanning the NIST IR lifecycle: a SYN flood DoS report, a ransomware incident documented with the 5 W's, and an HTTP-based malware-delivery investigation that traces back to a brute-forced admin account.

Track	Google Cybersecurity Professional Certificate
Course	Detection and Response
Framework	NIST Incident Response Lifecycle
Analyst	Chayan Panchal

1 Report 1 — SYN Flood DoS Attack

Attack type: Denial of Service — SYN flood. The web server stopped responding after being overloaded with SYN packet requests, consistent with a SYN flood.



How the Attack Broke the Handshake

1. NORMAL HANDSHAKE

Client SYN -> Server SYN-ACK -> Client ACK

2. FLOOD BEGINS

Attacker sends massive volume of SYN packets

3. RESOURCES EXHAUSTED

Server reserves resources per SYN, final ACK never arrives

4. LEGITIMATE USERS TIME OUT

No capacity left for real connection requests

NOTE The tcpdump/Wireshark logs confirmed the server was overwhelmed and unable to process further SYN requests from real visitors — full detail on the packet-level evidence for this attack is in the Wireshark Labs and tcpdump Labs folders.

2 Report 2 — Ransomware Incident (Incident Handler's Journal)

Field	Details
Date	July 23, 2024
Description	Documenting a ransomware cybersecurity incident
Tools used	None (documentation-only entry)

The 5 W's

Who	An organized group of unethical hackers
What	A ransomware security incident
Where	At a healthcare company
When	Tuesday, 9:00 a.m.
Why	Attackers gained access via a phishing attack, then deployed ransomware to encrypt critical files. Motivation appears financial — the ransom note demanded a large sum for the decryption key

OPEN QUESTIONS Two open questions the journal flags for follow-up: how could the healthcare company prevent a repeat of this, and should the company pay the ransom to retrieve the decryption key — both standard first-72-hours IR decisions.

3 Report 3 — Brute Force & HTTP Malware Delivery

Protocol: HTTP. Customers reported that visiting yummyrecipesforme.com prompted a download claiming to unlock new recipes; their computers slowed down afterward, and the site owner found themselves locked out of the admin account.

Investigating in a sandbox (to avoid impacting the company network), the analyst replicated the user experience with tcpdump running: downloaded the file, ran it, and watched the browser redirect to a fake lookalike site — greatrecipesforme.com.

tcpdump log findings

Browser resolved yummyrecipesforme.com via DNS, then established an HTTP connection and downloaded the malicious file

Traffic then suddenly redirected to a new IP for greatrecipesforme.com

Source review found injected code prompting users to download a file disguised as a browser update

ROOT CAUSE The attacker most likely used a brute force attack to compromise the admin account via a default password, then modified the site to serve the malicious redirect.

Recommended remediations

Disallow reuse of previous/default passwords — closes the vector that enabled the initial compromise

Require more frequent password updates — shrinks the window a compromised password stays useful

Implement Two-Factor Authentication (2FA) — a password alone is no longer sufficient for access

4 Quick Reference



Incident response — quick reference

NIST IR Lifecycle: Preparation -> Detection & Analysis -> Containment, Eradication
& Recovery -> Post-Incident Activity

5 W's documentation: Who / What / Where / When / Why

SYN flood tell: many SYN, no completed ACK, server resource exhaustion

Malware delivery: DNS lookup -> HTTP GET (download) -> unexpected domain pivot